

Grammaticality de-idealized

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Abstract

This paper introduces the Morphosyntactic-Value Model of Grammaticality (MVMG), a framework for (un)grammaticality distinct from grammars focused solely on language production or description. The model treats (un)grammaticality as conditioned stability in form–value relations accepted or rejected within communicative situations, integrating mapping viability, operator-value coherence, population licensing, and speaker-level feelings of anomaly. Previous approaches, including generative grammar, Construction Grammar, and psycholinguistic models, haven’t adequately explained why semantically coherent constructions may nonetheless be deemed ungrammatical, why certain transparent structures remain systematically unentrenched, or why categorical grammaticality clusters where it does. The present framework addresses these gaps by deriving categoricity from conventionalized norms, opportunity structure, and preemption dynamics within communicative situations, rather than from structural membership or intuition alone.

INTRODUCTION

Every competent speaker of English knows that **Can the have running* is impossible, but the source of this certainty proves remarkably elusive. What does it mean to say a sentence is ungrammatical? Consider these examples:

- (1) a. **Can the have running?*
- b. *Colorless green ideas sleep furiously.* (Chomsky, 1957)
- c. **I've finished it yesterday.*
- d. *?I saw Joan, a friend of whose was visiting.*
- e. *The bread the baker the apprentice helped made is delicious.*
- f. A: *How old are you?* B: **I have 25 years.*
- g. **Which did you buy car?*

While all might receive asterisks in many analyses, they represent fundamentally different types of unacceptability. Some constructions, like (1a), fail to establish any form–value relationship in English. Others, like (1c), involve a clash between the time semantics of clause tense and the word *yesterday*. Still others, such as (1d), show gradient or indeterminate status. (1e) illustrates what might be called theoretically predicted acceptability: constructions that linguistic theory predicts should be grammatical but that are consistently rejected by native speakers in ordinary judgment tasks. Still others, like (1f), are transparent and grammatical in related languages but lack licensing in English age predication. And a few, like the left-branch extraction of (1g), seem to be ruled out entirely, despite being apparently short and interpretable.

Different theoretical traditions have emphasized different aspects of grammaticality: formal approaches focus on abstract principles, usage-based theories stress frequency and entrenchment, and processing accounts highlight cognitive constraints. But the field still lacks a unified framework that can explain both categorical constraints and gradient acceptability while accounting for cross-linguistic variation and change over time.

This paper presents such a framework, building on insights from both generative and functional traditions to propose that grammaticality should be understood in terms of form–value relations that evolve within specific communicative situations. These relations interact with both universal processing constraints and sociolinguistic factors to produce the complex patterns of acceptability observed in actual language use. The framework rests on three key premises:

1. At its core, grammaticality involves conventional form–value relations within specific communicative situations (encompassing communities, dialects, and registers).
2. These relations interact systematically with processing constraints, sociolinguistic factors, and other linguistic subsystems (phonology, semantics, pragmatics).
3. Grammaticality reflects the degree to which an utterance establishes stable, conventionalized form–value relations within the relevant communicative situation. Various factors can disrupt this stability – from complete absence of a licensable analysis to conflicts between competing construals to lack of situational licensing – producing a spectrum of ungrammaticality effects unified by form–value instability.

These premises directly address key limitations in previous approaches to grammaticality. The first premise moves beyond the abstract competence model of early generative grammar while build-

ing on its analyses of systematic constraints. It provides a theoretical foundation for explaining both categorical constraints like (1g) and gradient acceptability like (1d), aligning with Francis's (Francis, 2022) arguments for incorporating gradience into core grammatical theory. The second premise resolves a long-standing tension between formal syntactic approaches and processing-based accounts by explicitly modeling their interaction rather than treating them as competing explanations. This explains why some ungrammatical constructions resist improvement through exposure while others become more acceptable with familiarity. The third premise provides a principled basis for phenomena that have resisted unified explanation in both generative and functional frameworks.

Section 1 examines the impasse in grammaticality theory, tracing how different theoretical traditions – from early generative grammar through contemporary experimental approaches – have attempted to reconcile form, meaning, and acceptability. Section 2 presents the framework in detail, defining the three constitutive variables that determine grammaticality and identifying recurrent diagnostic instability modes that govern judgments. Section 3 presents a simple formal model that illustrates how the framework's key components interact to produce grammaticality judgments. Section 5 explores the theoretical implications of this approach, particularly its relationship to generative grammar, Construction Grammar, and usage-based theories, and includes specific predictions about cross-linguistic variation in grammaticality judgments and the conditions under which satiation effects should occur. The paper concludes by acknowledging limitations of the current framework and suggesting directions for future research using corpus analysis, experimental methods, and cross-linguistic investigation.

1 THE IMPASSE IN GRAMMATICALITY THEORY

The concept of grammaticality remains elusive despite its centrality to linguistic theory. After decades of research, the field still lacks a comprehensive account of what makes an utterance grammatical or ungrammatical. This theoretical impasse stems from trying to treat grammaticality as a unified phenomenon, when it actually arises from the interaction of three distinct sources of failure. To understand why previous approaches have struggled, it is necessary to disentangle three components: structural viability, interpretive coherence, and situational licensing.

1.1 THE PROBLEM OF STRUCTURAL VIABILITY

The first challenge in defining grammaticality is distinguishing between structures that fail because they are fundamentally structurally unsound and those that fail for other reasons. Chomsky (1957), building on formal production systems developed by Post (1943), treated grammaticality as a categorical property defined by membership in a set of well-formed strings. This approach captures the intuition that some strings, like (1a), resist structural analysis entirely:

(2) **Can the have running?*

Here, the input crashes the structural analysis independent of meaning. The “categorical” view correctly identifies that a viable structural analysis is necessary for grammaticality. But by elevating this single condition to the *definition* of grammaticality, formal approaches struggled with evidence that speakers consistently provide gradient judgments for structures that are clearly analyzable but degraded. The competence-performance distinction (Chomsky, 1965) attempted to preserve categorical grammar by attributing gradience to processing limitations. But as Schütze (2016, p. 71) notes, this often leads to circularity, where problematic results are dismissed as performance artifacts.

The MVMG framework resolves this by recognizing that while structural viability (the successful mapping of form to category) is a necessary, categorical prerequisite, it isn't the sole determinant of grammaticality.

1.2 THE PROBLEM OF INTERPRETIVE COHERENCE

The second tension arises from the interplay of form and meaning. Chomsky's famous example (1b) shows that operator-valued form–value relations can be intact even when compositional semantics fails: the sentence is structurally a declarative, the subject is correctly identified both positionally and through agreement, and the tense suggests a general claim. What breaks is the compositional semantic relation: the lexical values don't cohere. Conversely, (1c) **I've finished it yesterday* shows compositional semantics can be transparent (the intended meaning is clear) while an operator-valued form–value relation fails. Here, the clash is between the present perfect's temporal value and the adverb's deictic anchoring.

Generative semanticists like Lakoff (1971) and McCawley (1968) demonstrated early on that many constraints have semantic motivations. Construction Grammar (Goldberg, 1995) has further shown how constructions carry meanings that interact with lexical semantics. Novel uses that align with established constructional meanings (like *She texted him the address*) are accepted, while those that clash (like **She disappeared him the evidence*) are rejected. A complete theory therefore needs to account for interpretive coherence: the degree to which a structurally viable form yields a stable, non-contradictory interpretation.

1.3 THE PROBLEM OF SITUATIONAL LICENSING

Finally, even structurally viable and semantically coherent forms may be ungrammatical if they violate community conventions. Sociolinguistic research (Labov, 1972) demonstrates that grammaticality references community norms. Cross-linguistic variation shows that each language community conventionalizes particular form–value mappings through historical processes. These conventions become entrenched through statistical preemption (Goldberg, 2011) – speakers learn that certain forms are ungrammatical precisely because they encounter alternative expressions in contexts where the ungrammatical form would otherwise be expected.

Consider (1f), repeated here:

- (3) A: *How old are you?* B: **I have 25 years.*

This utterance is structurally viable and semantically transparent (the intended meaning is clear), but it is categorically rejected in English. The same form is grammatical in French (*J'ai 25 ans*) and Spanish (*Tengo 25 años*), demonstrating that the constraint is community-specific. Usage-based approaches (J. Bybee, 2006) emphasize that such rejection arises from the absence of situational licensing. The English-speaking community has conventionalized a different way to express age, leaving this form without positive evidence.

These three problems – structural viability, interpretive coherence, and situational licensing – correspond exactly to the three constitutive variables of the MVMG framework presented in Section 2. Acknowledging them as distinct but interacting components makes it possible to move beyond the impasse of trying to force all grammaticality judgments into a single “competence” definition or a purely probabilistic usage model.

2 THE MORPHOSYNTACTIC-VALUE MODEL OF GRAMMATICALITY

This section presents the proposed MVMG framework. The term *VALUE* is used here in the Saussurean sense of *valeur* (Saussure, 1916): the identity of a linguistic unit is constituted not by intrinsic properties but by its position in a system of contrasts, what it patterns with, what it opposes, and what interpretations it makes available. It is not to be confused with feature-value assignment or decision-theoretic value. Throughout, I treat grammatical knowledge as a conditioned form–value relation; when this relation is sufficiently stable in a communicative situation, with a dominant value reliably recoverable and socially licensed, I refer to it informally as an established form–value relationship.

The formal architecture rests on three constitutive variables: mapping viability (*map*), operator-value coherence (*K*), and situational licensing (*C_t*). Their interaction determines grammaticality (§3). Table 1 summarizes these variables; the formal definitions appear in §3.

Variable	Component
$\text{map}(u, c)$	Mapping Viability (Categorical prerequisite)
$K(u, c)$	Operator-Value Coherence
$C_t(u, c)$	Situational Licensing (Social convention)

Table 1: The three constitutive variables of the MVMG framework.

For expository purposes, it is useful to distinguish recurrent *INSTABILITY MODES*: diagnostic categories that correspond to different ways of driving *map*, *K*, or *C_t* toward zero, plus processing and prescriptive factors that modulate how grammatical something *feels* without affecting grammatical status directly.

2.1 DIAGNOSTIC CATEGORIES

The diagnostic categories described below aren’t primitives of the model but rather convenient labels for the most common routes to ungrammaticality. Each can be traced to one or more of the constitutive variables, as the formal core in §3 makes explicit.

2.1.1 Operator-valued form–value relations within communicative situations

At its foundation, grammaticality depends on stable relationships between linguistic forms and the operator values they realize within specific communicative situations. That form–value relations exist at all levels of grammatical organization, from morphemes to abstract syntactic patterns, is a core insight of Construction Grammar (Goldberg, 1995). MVMG narrows the target of categorical grammaticality to role-functional contrasts: closed-paradigm, high-opportunity, update-critical resources that configure how an utterance enters public coordination. A companion paper develops this operator-stratum account in detail and argues that the boundary is functional rather than substance-based (Reynolds, 2026).

Operator exponents, like words, are typically polysemous: a single form participates in multiple form–value relationships, usually with one dominant value (Kilgariff, 2004). The English past tense, for instance, usually denotes past time, but it can also mark social distance (*Could you help me?*) or low likelihood (*If I went tomorrow...*). What matters for grammaticality is whether a stable relationship exists within the relevant communicative situation, not whether the form has a single fixed meaning.

Neuroimaging work provides converging evidence for this integration of form and meaning at the processing level. Fedorenko and colleagues find that brain regions activated by syntactic manipulations overlap substantially with those activated by semantic ones (Fedorenko et al., 2011, 2012, 2024), suggesting partly shared processing resources rather than wholly separate modules.

Following Wiese (2023, p. 3), grammaticality is conditioned by regularities within a “communicative situation”. The emergence claim defended below is narrower: categoricity emerges from the licensing dynamics in §4.4, not from an unanalyzed appeal to usage.

Communicative situations aren’t static or mutually exclusive. A speaker navigates overlapping contexts that shift on multiple dimensions: the immediate situation (formal meeting vs. casual chat), durable social positions (regional, generational), and the norms they’re orienting to (which can shift with audience). This fluidity explains style-shifting, dialect formation, and the gradual acceptance of initially marginal constructions. What matters for grammaticality is the stability of form–value relationships within the relevant communicative situation.

This is visible in early child language, where toddlers produce utterances that deviate from adult norms but remain internally consistent within the child’s developing system. A toddler in a monolingual English household might say:

- (4) *Ava cookie.* (intended as ‘Ava = I want a cookie’)

Although this differs from adult English norms, it may not be perceived as ungrammatical by the child’s regular caregivers. Used with the same pragmasemantic force among anglophone adults, the construction would be judged ungrammatical.

Multiple modal constructions provide another clear example of variety-relative grammaticality:

- (5) *I might could help you with that.*

This combination of modal auxiliaries is systematically possible for some American English speakers, who can productively generate similar constructions (Morin & Grieve, 2024). Speakers from communities where only single modals are grammatical, though, typically reject such combinations as ungrammatical.

A similar dynamic appears in code-mixing among bilingual speakers, where combinations of forms from different languages can be grammatical within that bilingual community’s norms but not without. Consider a Spanish-English bilingual speaker who uses a Spanish progressive auxiliary with an English lexical verb:

- (6) *Ayer, estábamos lifting en el gym durante una hora.*
 yesterday be.IMPF-1PL lifting in the gym for an hour
 ‘Yesterday, we were lifting in the gym for an hour.’

Within the right communicative situation, this utterance is grammatical. The Spanish auxiliary *estábamos* combines with an English participial form *lifting* to form the progressive aspect. This cross-linguistic combination of morphology and a lexical verb is consistent with local norms, where code-mixed utterances are common and meaningful. In contrast, a standard monolingual Spanish community, which expects fully Spanish progressive structures (*estábamos levantando pesas*), may judge the example in (6) as ungrammatical. The use of intransitive *lifting*, specific to the gym community, further illustrates just how localized grammaticality judgments can be.

This doesn't mean bilingual communities simply accept any combination of languages. As Toribio (2001) reports, Spanish–English bilingual speakers judge examples like (7) as unacceptable, showing that even in bilingual communities, there are systematic constraints on which language combinations are permitted.

- (7) * *Los enanitos intentaron pero no succeeded in awakening Snow White* (Toribio, 2001)
 the dwarfs try.PST-3PL but not succeeded in awakening Snow White
 'The dwarfs tried but did not succeed in awakening Snow White.'

In a slightly different case, as a second-language speaker of Japanese, I used to say

- (8) * *Kawaii da.*
 cute-PRES COP.PRES
 '(That)'s cute.' (intended)

Conventionally, though, the redundant tense marking has no accepted meaning, making the use of the copula ungrammatical to most Japanese speakers, though it felt meaningful and grammatical to me. The example underscores that stable form–value relations emerge from and depend on the shared linguistic routines of a particular community, and individuals with differing trajectories of acquisition may diverge in their grammatical judgments.

The specific linguistic context can matter too:

To take an obvious case which Jerry Morgan [(1973)] discussed recently, certain combinations of words are extremely strange if presented in isolation but are perfectly normal as answers to certain questions. *Spiro conjectures Ex-Lax* would generally be felt to be unintelligible if presented out of context but is a perfectly normal answer to the question *Does anyone know what Mrs. Nixon frosts her cakes with?* (McCawley, 1974, 252, italics added)¹

These examples illustrate that grammaticality is communicative-situation-relative. Conversants who differ in immediate situation, social position, or the norms they orient to may disagree on whether a form is grammatical – and both may be right within their respective contexts. The linguist's task is therefore to determine whether a form is grammatical from any position in a communicative situation, or systematically excluded from all.

The stability of form–value relations within speech communities can be empirically modeled, as demonstrated by Blythe and Croft (2009). Their utterance selection model treats speech communities as networks where speakers track and reproduce linguistic variants based on their interactions. When applied to dialect formation, these models show how competing linguistic variants spread and eventually stabilize, with initial variant frequency strongly predicting which form will prevail.

Different language communities encode different distinctions as grammatically obligatory, reflecting what each community deems relevant enough to systematically mark. Over time, communities establish patterns – grammatical constructions – that reliably signal these chosen distinctions. As a result, what counts as a grammatical necessity in one language may be optional or absent in another.

The progressive aspect provides a useful example. In Standard English, it's not merely an option but an obligatory grammatical marker for ongoing, incomplete actions:

¹ The humour derives from political tensions of the Watergate era, when both Vice-President Spiro Agnew and President Nixon would ultimately resign from office.

(9) *She is studying right now.*

Here, the progressive form *is studying* isn't just a stylistic choice. It's the recognized, grammatical way to express a currently unfolding activity. English speakers strongly prefer (and in many contexts, demand) the progressive construction to convey immediacy and ongoingness.

French, by contrast, doesn't treat the progressive aspect as a grammatically mandatory distinction. While one can signal ongoing activity through adverbs or periphrastic constructions, standard French doesn't have a dedicated progressive form. The sentence:

(10) *Elle étudie maintenant.*
'she studies/is studying now'

can comfortably describe a currently ongoing action without any need for special morphology. The community hasn't defined this aspectual distinction as an obligatory operator contrast. What is grammatically necessary in English – employing the progressive to signal ongoingness – is simply not a requirement in French.

A similar dynamic appears with evidentiality, the grammatical marking of information sources. In Turkish, evidentiality is systematically encoded through verb forms and particles that distinguish between directly witnessed events and those inferred or reported:

(11) *Gelmiş*
'He/she came (apparently)'
(i.e. the speaker wasn't a witness but inferred or heard about it.)

The language community treats evidential distinctions as central enough to be baked into the grammar. A speaker can't simply omit evidential marking without sounding ungrammatical.

Of course, while English speakers can say *I heard that he arrived* or *He must have arrived*, these are optional lexical or modal resources rather than required elements of the grammar. The English community simply doesn't regard evidential distinctions as something that has to be encoded as an obligatory operator contrast.

Kilani-Schoch and Dressler (2005) demonstrate this principle systematically in their analysis of verbal morphology across French and other Romance languages. They show how apparently similar verbal systems can encode quite different semantic distinctions as grammatically obligatory, reflecting each community's conventions about which meaning distinctions have to be systematically marked. For instance, while both French and Italian mark aspect morphologically, they differ in which aspectual distinctions are grammaticalized versus left to optional lexical expression.

Establishing a form–value mapping is a prerequisite for grammaticality, but it isn't sufficient. Individuals routinely interpret strings that are nevertheless judged ill-formed within the relevant communicative situation. That very flexibility forces the grammar to police *how* a relation is licensed, not merely *whether* one can be imagined in principle. In other words, additional factors accumulate evidence against candidate mappings: (i) conflict with conventional constructional values contributes negative evidence (§2.1.2), (ii) high recovery cost down-weights the mapping (§2.1.3), and (iii) absence of conventionalization leaves the mapping with insufficient positive evidence (§2.1.6). Only by integrating these weighted contributions does the framework predict the empirical fact that **Furiously sleep ideas green colorless* elicits universal rejection while (1e) feels bad in everyday quotation yet

is accepted as “grammatical but hard to process” once speakers are walked through the intended structure. The combined weighting therefore concentrates probability mass on the much smaller subset of imaginable relations recognized as bona fide grammatical resources.

One might object that the community’s verdict alone should settle the matter: if speakers converge on using a form, why not define grammaticality as simply “whatever the community accepts”? That move is circular and empirically empty: it can neither flag incipient innovations whose status is still contested nor explain why stable dialectal differences persist despite mutual intelligibility. A purely sociological definition would also mis-classify well-attested performance errors (e.g. agreement slips in live speech) as “grammatical” whenever they pass unnoticed, and it would leave us with no basis for diagnosing why learners systematically avoid forms that the target community fully endorses. Separating the mapping, compatibility, and acceptance layers therefore avoids trivializing the concept of grammaticality while still granting the community the final say on which relationships endure.

2.1.2 Operator-value compatibility

Beyond the existence of form–value relations, grammaticality requires compatibility among the operator values those relations encode. This component captures cases where individual elements are well formed but their combination creates conflicts among grammaticalized contrasts activated in the communicative situation.

Consider the temporal incompatibility in:

(12) * *I’ve finished it yesterday.*

Here, the present perfect construction encodes current relevance while *yesterday* specifies completed past time. Unlike lexically incongruous combinations like *colorless green ideas*, which remain grammatical because the clash doesn’t mis-set an operator value, this example shows a direct conflict between the temporal value encoded by the grammatical construction and the lexical temporal specification.

The age example belongs elsewhere:

(13) * *I have 25 years.* intended as ‘I’m 25 years old’

In English, *have+years* is a licensed frame for duration, remaining time, or experience, but ordinary age predication is licensed through a different frame. The intended value is transparent, so this is not a K failure in the narrow sense. It is a licensing failure for an age-predication construction in English, even though cognate-looking frames are licensed in French and Spanish.

Sometimes value incompatibility involves conflicting information-structural requirements:

(14) * *Who did the lifeguard who saved _ work in New Jersey?* (Cuneo & Goldberg, 2023)

Here there is a clash in information structure: the same participant is simultaneously focused through fronting *who* while being backgrounded by the relative clause construction (Cuneo & Goldberg, 2023).

These examples also delimit *K*. The live ϕ_j in the energy model are constraints on grammaticalized, operator-like features in *c*: tense–aspect anchoring where tense–aspect is obligatory, argument-linking requirements where a construction licenses roles, information-structural requirements where a construction grammatically packages focus or backgrounding, and indexical contrasts only when they have entered a closed accountable repertoire. This makes *K* partly derivative of licensing facts

at the feature level. That dependence is a virtue, because the model should not treat arbitrary lexical implausibility as grammatical incoherence.

2.1.3 Processing constraints

Language processing engages both dedicated language networks and domain-general cognitive systems (Fedorenko et al., 2024). When constructions overload these systems – particularly through multiple long-distance dependencies or heavy embedding – they may trigger feelings of ungrammaticality despite being structurally well-formed.

(15) *The bread the baker the apprentice helped made is delicious.*

Evidence for these constraints comes from multiple sources. Studies of sentence processing show that dependencies spanning multiple intervening elements increase processing difficulty, with new referents between dependent elements compounding memory load (Gibson, 2000, 2026). In (15), while each individual relation (like *the apprentice helped* and *the baker made*) is interpretable in isolation, their nested combination overwhelms incremental processing. The construction is grammatical, but excessive bridging costs from its nested structure trigger feelings of ungrammaticality.

Rather than reflecting simple memory limitations, processing constraints emerge from the interaction between specialized language networks and other cognitive systems (Fedorenko et al., 2024). On this processing-level interpretation, what speakers experience as difficulty may reflect demands distributed across specialized brain systems rather than a single cognitive bottleneck.

The diachronic claim requires a further bridge through speaker behaviour and transmission. Forms that impose lower processing demands are easier to store, recall, and reuse. When speakers preferentially select these more manageable structures, simpler patterns can spread through the linguistic community and eventually become conventional. This process can be seen at work when languages simplify nested relative clauses or reduce complex morphological paradigms. The processing evidence therefore supports a plausibility claim about one pathway into conventional stability, not a direct neural explanation of grammatical change.

Processing constraints also interact with other components. A construction that is marginal due to low situational acceptance may become completely unacceptable when combined with processing difficulty. Conversely, highly entrenched constructions may remain acceptable despite considerable processing demands, suggesting that strong conventionalization can partially offset processing costs.

2.1.4 Dependency locality

Integration cost rises with dependency length; self-paced reading shows difficulty peaks at loci of long-distance integration, with locality accounting for 59–73% of variance in reading times (Grodner & Gibson, 2005). Because locality effects are so robust and well-documented, the formal treatment in §3 treats dependency locality as a dedicated component rather than hiding it inside a residual term.

Locality influences both the subjective feeling of ungrammaticality and the licensing dynamics. High-locality constructions feel worse to process, even when structurally well-formed. And because such constructions are harder to produce and parse, speakers tend to avoid them, which over time reduces their conventionalization. The formal treatment appears in §3.

2.1.5 Socio-pragmatic indexicality

The value of a construction extends beyond semantic content to include socio-pragmatic dimensions – how linguistic forms index aspects of social context, speaker identity, group membership,

stance, or interpersonal relationships (Eckert, 2012; Silverstein, 1976).

In many varieties of Latin American Spanish, for example (e.g. the Río de la Plata region), speakers use *vos* and its associated verb forms instead of the *tú* forms used elsewhere in the Spanish-speaking world (Bertolotti, 2016):

- (16) ¿*Vos querés un café?*
 you.SG want.2SG-VOS a coffee
 ‘Do you want a coffee?’

Here, the use of *vos* rather than *tú* not only denotes the second-person singular hearer – the person being offered a coffee – but also indexes the speaker’s regional identity and familiarity with the local dialect. This indexical value may convey closeness, solidarity, or membership in a particular geographic and social community.

This situational view of grammaticality can manifest asymmetrically. Speakers from the Río de la Plata region may view a conversational situation as accommodating both their own norms and those of *tú*-using interlocutors – the situation itself can encompass both *vos* and *tú* as grammatical options. But speakers from *tú*-only regions might conceptualize the same situation more restrictively, defining it in a way that categorically excludes *vos* as a grammatical possibility. This asymmetry doesn’t depend on different understandings of the forms themselves, but can arise from different ways of defining what the communicative situation allows, influenced by the indexical values attached to *vos* versus *tú* in their respective communities.

The indexical value of constructions extends to phonology, but as long as there is no conflict with operator value, grammaticality isn’t at question. Babel (2025) provides an example. In a study conducted in Bolivia, participants were presented with audio stimuli where only the vowels were manipulated to reflect either a highland or lowland accent. This presented participants with incongruent identity cues (e.g. vowels from one accent along with consonants from the other). But this didn’t trigger feelings of ungrammaticality. Instead, vowel contrasts activated expectations about consonant features and discourse markers, resulting in some participants’ “hallucinating” identity-linked features that weren’t present in the signal.

This demonstrates that the value of a construction goes beyond semantic features to include socio-pragmatic aspects that shape how speakers and hearers negotiate authority, identity, solidarity, and other interpersonal relations. Recognizing these indexical dimensions is essential for understanding why certain forms feel natural and grammatical to some speakers and out of place or even ungrammatical to others.

2.1.6 Situational entrenchment and acceptance

Even well-formed constructions with clear meanings may be judged ungrammatical if they lack conventionalized acceptance. This component captures the degree to which a form–value relation has become entrenched within relevant communicative situations.

- (17) * *We sheared three sheeps.*

The regular plural marking is semantically transparent and structurally parallel to other English plurals, but the irregular form *sheep* is entrenched across English-speaking communicative situations, making the regularized version unacceptable. Without a metaphorical or playful justification (as in

the black sheeps of the family, where the irregular plural might signal a figurative usage), the utterance is ungrammatical.

Situational entrenchment operates independently from the other components. A construction may have perfect value compatibility and minimal processing demands but still be rejected because a different form has been conventionalized for that meaning. This is particularly evident in cases of extreme rarity. Some constructions are so infrequent that speakers lack a shared consensus about their status.

The independent relative genitive pronoun *whose* provides an example, being so unusual that Hankamer and Postal (1973) deem it non-existent. The contexts that license this construction require the simultaneous convergence of distinct pragmatic and syntactic conditions – sufficient accessibility of both possessor and possessum, the appropriate information structure, and an environment allowing ellipsis – which are seldom met all at once (Reynolds under review):

(18) [?] *I saw Joan, a friend of whose was visiting.*

(adapted from Huddleston & Pullum 2002, p. 472)

A search of the 1-billion-word Corpus of Contemporary American English returned no instances of this construction. Crucially, though, the opportunity set is also tiny: contexts requiring an independent relative genitive are vanishingly rare. Speakers therefore have little evidence either way – neither positive tokens nor the systematic absence that would accumulate if the construction were regularly passed over. The result is high *uncertainty* rather than confident rejection. Some speakers can make the analogical leap from similar constructions to accept examples like (18); others remain unsure; still others, lacking sufficient exposure, can't construct a stable form–value relation at all. Even among those who grasp the construction analytically, Shain et al. (2020)'s fMRI research suggests that its unexpectedness would trigger high surprisal, compounding the uncertainty with processing difficulty.

This contrasts sharply with stable gaps like left-branch extraction (§2.1.7), where the opportunity set is large and the construction is systematically untaken – giving speakers overwhelming evidence that it isn't licensed.

2.1.7 Apparent categorical gaps

Some constructions face rejection so persistent that it *appears* categorical. Left-branch extraction is the textbook case:

(19) * *Which did you buy [__ car]?*

The intended meaning is easily grasped; nothing in the semantics or pragmatics prevents understanding the speaker's intent. Nor does the construction seem excessively complex to process. Yet English speakers categorically reject it, and Snyder (2022, p. 22) treats this kind of pattern as a core case of “ungrammaticality”.

Under MVMG, such patterns are analyzed not as hard representational bans but as stable gaps: situational licensing is driven toward zero by persistent preemption. Because the communicative job is common and an established competitor always wins (e.g., *Which car did you buy?*), speakers accumulate overwhelming evidence that the construction isn't part of the repertoire. Additional processing penalties – such as systematic garden-path reanalysis when the bare *wh*-phrase is encountered –

can strengthen the subjective categoricity without any independent structural veto being posited. The formal dynamics appear in §3.

Several diagnostic criteria help identify these stable-gap constructions:

1. Persistent unacceptability: The construction never gains acceptance; familiarity and repetition don't shift judgments.
2. Categorical rejection: Speakers find the constructions simply impossible, with no intermediate ratings.
3. Resistance to satiation: Unlike processing-heavy cases, repeated exposure doesn't improve acceptability.

Prediction: If participants are led to treat presented tokens (like LBE) as intentional dialectal or ingroup resources rather than errors, then satiation (or at least reduced feeling of ungrammaticality) should be more likely than in standard paradigms where the construction is treated as simply wrong.

If future work uncovers constructions that resist even this stable-gap analysis – forms for which no structural analysis is available at all – they would represent a residual class of hard representational bans. For now, treating classic cases like LBE as driven to zero under preemption keeps the constitutive core minimal and pushes the explanatory work into the dynamics of situational licensing.

2.2 DIAGNOSING (UN)GRAMMATICALITY AT A GLANCE

For exposition the recurrent instability modes can be read as a short decision tree (the formal definitions appear in §3):

Condition	→ Canonical outcome
no viable structural analysis	→ NONSENSE (<i>*Can the have running</i>)
semantic/indexical clash	→ value incompatibility (<i>*I've finished it yesterday</i>)
rarely encountered, low certainty	→ community-novel (<i>friend of whose</i>)
systematically preempted	→ stable gap (left-branch extraction)
high processing cost, structure recoverable	→ transient ill-formedness (centre embedding)
otherwise	→ grammatical

2.3 PATTERNS OF (UN)GRAMMATICALITY

When these recurring components interact, they produce systematic patterns in how constructions succeed or fail to achieve grammatical status. Understanding these patterns helps explain the diverse phenomena traditionally grouped under “ungrammaticality”

2.3.1 When constructions are ungrammatical

Ungrammaticality arises when expected patterns within the relevant communicative situation are violated in ways that can't be reconciled through available interpretive strategies. The diagnostic categories generate distinct types of violation:

NO VIABLE FORM-VALUE RELATION. In some cases, the form of an utterance simply doesn't map onto any conceivable interpretation:

(20) **Can the have running?*

Here, the modal *can* expects a subject and a verb phrase to form a coherent proposition. Instead, *the have running* neither yields a noun phrase nor a legitimate verbal structure. The result is a form for which no stable meaning is available. With no recognizable pattern to anchor on, the utterance remains nonsensical and ungrammatical.

VALUE INCOMPATIBILITY. When the operator values encoded by grammaticalized elements clash, the construction fails despite having well-formed individual components. This includes temporal conflicts (as in **I've finished it yesterday*) and information-structural contradictions (as in the lifeguard example).

PROCESSING OVERLOAD. Constructions that exceed cognitive processing capacity trigger ungrammaticality judgments even when structurally well-formed. Multiple center embeddings exemplify this pattern, where incremental processing becomes impossible despite theoretical grammaticality.

LACK OF SITUATIONAL ENTRENCHMENT. Even transparent, processable constructions fail when they violate established conventions. The *sheeps* example shows how regularization can be preempted by entrenched irregular forms; **I have 25 years* fails as ordinary English age predication because the relevant frame is not licensed for that communicative job. The extreme rarity of independent relative *whose* prevents stable conventionalization.

APPARENT CATEGORICAL GAPS. Some constructions are rejected so persistently that they behave as if categorically banned. Under MVMG, these are analyzed as **STABLE GAPS**: the structure is viable and interpretation coherent, but situational licensing is driven toward zero by persistent preemption in a large opportunity set (§4.3, §4.5). Left-branch extraction is the textbook case.

2.3.2 Degrees of violation

Grammaticality isn't binary; rather, it reflects the scope and intensity of component mismatches. Violations can range from mild, easily recoverable deviations to complete breakdowns in interpretability.

Minor violations might involve a single component with partial conflict – for instance, a construction with moderate processing difficulty but full value compatibility and strong situational entrenchment. Such cases often receive intermediate acceptability ratings and may improve with exposure.

Severe violations typically involve multiple components or complete failure of a single critical component. A construction lacking any viable form–value relation represents total breakdown, while violations of categorical constraints produce consistent, strong rejection regardless of other factors.

The interaction between components can amplify or mitigate violations. Strong situational entrenchment can partially offset processing difficulty, while semantic transparency can make marginal syntactic patterns more acceptable. Conversely, multiple minor violations can compound to produce strong ungrammaticality judgments.

2.4 THE SUBJECTIVE EXPERIENCE OF GRAMMATICALITY

While the constitutive variables determine objective grammaticality, speakers experience violations through subjective feelings and judgments that don't always align perfectly with grammatical status.²

²The distinction between objective grammaticality and the subjective feeling has a philosophical analogue in Cavell's (1969) distinction between knowledge and acknowledgment. For Cavell, acknowledgment isn't a separate capacity from knowing but an inflection of it – bearing knowledge toward the situation that produced it. Speakers don't merely *know* that a string is ungrammatical; they *register* it as a violation, with metacognitive consequences. This matters for

2.4.1 The feeling of ungrammaticality

The subjective **FEELING OF UNGRAMMATICALITY** is speakers' metacognitive response to linguistic violations. This distinction parallels other well-studied metacognitive feelings like the **FEELING OF KNOWING** (FOK; Hart, 1965), which arises when one feels certain of knowing something but can't retrieve it.

Note that there is no positive feeling of grammaticality, just as there is no feeling of having sufficient oxygen, only negative feelings experienced in the absences. The feeling of ungrammaticality, then, can be seen as the negative response triggered when an utterance violates expected form–value patterns.

This notion echoes Edward Sapir's concept of “form-feeling” – an often unconscious grasp of language patterns (Sapir, 1921, 1927). Evidence from aphasia supports the distinction between grammaticality and its subjective experience. Patients with Broca's aphasia, who produce agrammatic speech, often exhibit self-monitoring behavior, attempting self-correction and expressing frustration with their grammatical errors (Oomen et al., 2005).

The feeling of ungrammaticality can be conceptualized as a negative response speakers experience when unstable or missing form–value relations are detected during processing. This response wouldn't define grammaticality itself but would instead serve as a speaker-level heuristic for detection. This distinction helps explain:

1. Gradient Judgments: Varying degrees of certainty about marginal constructions reflect differences in the strength of the negative response rather than categorical grammatical status.
2. Satiation Effects: Repeated exposure to marginal constructions might not change grammatical status but could attenuate negative responses by increasing familiarity.
3. Cross-Linguistic Variation: Differences in how reliably operator-value violations are detected affect the consistency and intensity of negative responses.
4. Mismatch Between Intuition and Reality: Constructions can be objectively grammatical but trigger negative responses due to processing difficulty, or vice versa. That is (un)grammaticality can be illusory (Fanselow, 2021).

2.4.2 Distinguishing objective grammaticality from subjective ratings

A fundamental challenge in grammaticality research lies in distinguishing the objective property of grammaticality from the subjective feelings and ratings that speakers provide. This distinction proves crucial for both theoretical development and empirical investigation.

TWO LEVELS OF ANALYSIS.

The framework distinguishes two conceptually distinct levels:

This distinction parallels the measurement problem in other sciences: temperature can't be directly observed but has to be inferred through the behaviour of thermometers. Similarly, grammaticality itself remains a theoretical construct accessed through various behavioural measures, with acceptability ratings being just one imperfect window into the underlying phenomenon.

WHAT FACTOR ANALYSIS REVEALS.

When researchers conduct factor analyses on acceptability ratings – a common approach in experimental syntax – they necessarily target the structure of the subjective feeling, not objective gram-

LLM-based grammaticality research: LLMs can produce judgments that correlate with human ones, but whether they acknowledge the norms they're applying – or merely process patterns that correlate with those norms – remains an open question (Techio, 2026).

Table 2: Objective grammaticality versus subjective experience

Level	Nature	Observable through
Objective grammaticality	Whether a form–value relation is licensed in the communicative situation	Converging evidence: corpora, production, repair behaviour
Subjective feeling	Metacognitive warning signal that “something is wrong here”	Directly observable: ratings, eye-tracking, self-reports

maticality directly. The factors that emerge from such analyses reflect both genuine grammatical constraints and additional sources of variance that modulate subjective responses without affecting grammatical status.

- Operator licensing: When this factor loads heavily on rejected items and converges with corpus-rate-per-opportunity, elicited production, and repair behaviour, it reflects C_t .
- Value coherence: This factor, including information-structure clashes, belongs to K only when it tracks constraints on grammaticalized operator values.
- Situational entrenchment/frequency: High-frequency patterns that speakers have conventionalized provide evidence about C_t , but raw frequency has to be normalized by opportunity set.
- Sociolinguistic appropriateness: This factor has mixed status. Indexical clashes can affect C_t when they enter a closed accountable repertoire, but stylistic infelicities typically affect F .
- Processing cost/memory load: This factor belongs to F unless it feeds back into production, repair, and later licensing through the loop in §3.3.2.
- Morphophonological well-formedness: When phonological patterns realize obligatory operator exponents, violations affect K or C_t . Otherwise, they influence the subjective response.

This distinction between objective grammaticality and subjective feeling has immediate methodological consequences. If research aims to uncover the grammar itself, investigators should model factors only when they also appear in the independent indicators for C_t and K . Conversely, research into human judgment behaviour should retain the full factor structure, as it is precisely the subjective feeling that predicts how speakers will rate sentences presented out of context.

The distinction also explains certain puzzling phenomena in the experimental literature. Satiation effects may reflect changes in subjective feeling without necessarily altering objective grammaticality. A construction might remain objectively ungrammatical in the community’s system while triggering progressively weaker negative responses through familiarization. Similarly, the persistence of gradient judgments for certain phenomena suggests that subjective response operates on a continuum even when the underlying grammatical system makes categorical distinctions.

Most importantly, recognizing this distinction prevents circular reasoning in grammatical theory. Rather than defining grammaticality through acceptability ratings – which would conflate objective status with subjective response – the framework treats ratings as one type of evidence among many for inferring the underlying grammatical system. This approach maintains the empirical vulnerability of grammatical hypotheses while acknowledging the complex relationship between competence and performance, between the linguistic system of a communicative situation and individual speakers’

responses to it.

2.4.3 Misattribution effects

Sometimes the subjective feeling of ungrammaticality doesn't accurately reflect objective grammatical status. Both false positives (grammatical constructions feeling ungrammatical) and false negatives (ungrammatical constructions escaping detection) occur systematically.

WHEN GRAMMATICAL CONSTRUCTIONS FEEL UNGRAMMATICAL.

Listeners and readers can misanalyze utterances, triggering feelings of ungrammaticality even though the construction conforms to standard patterns:

(21) *The old man the boats.* (Ritchie & Thompson, 1984)

On first reading, one might treat *old man* as a noun phrase, resulting in nonsense. But the sentence actually has *the old* as the subject (meaning 'the elderly'), and *man* as a verb. Interpreted correctly, the sentence means 'The elderly operate the boats', and is fully grammatical.

Processing overload provides another source of misattribution. Heavily nested structures like (15) are grammatically well-formed but trigger negative responses due to cognitive limitations rather than grammatical violations.

WHEN UNGRAMMATICAL CONSTRUCTIONS ESCAPE DETECTION.

Conversely, objectively ungrammatical utterances may fail to trigger negative responses when semantic content is compelling:

(22) * *In Michigan and Minnesota, more people found Mr. Bush's ads negative than they did Mr. Kerry's.* (Pullum, 2009)

The intended interpretation is clear, and hearers readily infer the meaningful comparison. Syntactically, however, the structure means something else. The first clause suggests comparing numbers of people, but the second clause has *they* ('more people'), making the meaning 'more people did x than more people did y', which is nonsensical. But this violation often escapes detection.

Agreement errors can similarly hide within complex noun phrases:

(23) * *The patchwork of laws governing background checks, addressing assault-weapons limits, and regulating open-carry practices help explain why people continue to be wounded and killed.*
(modified from Corbett, 2016)

The singular subject *patchwork* clashes with plural *help*, but the intervening plural nouns mask this violation. Cognitive resources devoted to processing complexity can prevent speakers from registering the agreement error.

2.5 MECHANISMS OF GRAMMATICAL CHANGE

The stability of form–value relations isn't static. Various motivations drive the acceptance of new forms and the rejection of established ones, explaining how grammatical systems evolve over time.

2.5.1 Semantic motivations for reanalysis

Speakers regularly deploy metaphors, analogies, and context-induced reinterpretations that stretch or shift the meaning of existing forms. Over time, such re-analyses can yield new grammatical constructions that weren't previously part of the language's repertoire.

A well-documented example is the development of the *going to* futurate construction. Historically, *going to* described physical motion toward a place:

(24) *I am going to London.*

Frequent usage in contexts where motion implied subsequent action (e.g. *I am going to fetch some firewood*) led to semantic shift. The directional component was reinterpreted as marking future intention rather than spatial goals:

(25) *It's going to rain.*

What once expressed physical trajectory now functions as a predictive futurate marker. The form–value relation changed through semantic reanalysis, driven by communicative utility.

2.5.2 Social motivations for accepting or rejecting forms

Linguistic variants often serve as markers of regional background, class, ethnic identity, or age group. These social cues motivate speakers to favor some forms over others, leading to shifts in what counts as grammatical.

The decline of the simple past (*le passé simple*) in modern spoken French illustrates social motivation. Forms like *il alla* ('he went') became associated with formal, literary, or provincial speech. To avoid signaling undesirable social affiliations, speakers gravitated toward compound forms like *il est allé*, which lacked these status-laden connotations.

This transformation underlines how grammatical acceptance reflects changing social dynamics. Forms that were once fully grammatical can become socially marked and gradually excluded from normative grammar.

2.5.3 Structural motivations

Structural considerations arise from cognitive and communicative demands, including processing limitations, the need for clarity, and the influence of analogy.

PROCESSING CONSTRAINTS AND MEMORABILITY. Languages favor patterns that minimize processing demands. Forms requiring multiple long-distance dependencies or heavy embedding are less likely to achieve stable transmission across generations. This pressure toward processability shapes grammatical evolution, with simpler patterns spreading through communities.

ANALOGICAL EXTENSION. When speakers encounter new communicative challenges, they often extend known patterns to new contexts:

(26) *⁑ I asked about what did she do.*

Though currently marginal, such forms appear to gain traction by analogy with main-clause interrogatives (*What did she do?*). If accepted, this analogical extension would represent structural motivation reshaping the grammar.

2.5.4 Iconic motivations

Forms are sometimes accepted precisely because their structure reflects their meaning. Reduplication for intensity provides a simple example:

(27) *a big big problem*

The formal repetition iconically mirrors the magnitude being expressed. Such patterns are readily interpretable and likely to spread because the form–value link is immediately apparent.

Note that iconicity remains syntactically constrained. It occurs in pre-head modifiers but is ungrammatical as predicative complement:

(28) * *the problem was big big*

These diverse motivations – semantic, social, structural, and iconic – can interact and sometimes conflict. While Optimality Theory has productively modeled such competing pressures through ranked constraints (Prince & Smolensky, 2004), MVMG views their resolution as arising from usage within communicative situations rather than solely from fixed rankings. The specific weighting of motivations reflects historically established patterns and communicative needs.

3 A FORMAL CORE: GRAMMATICALITY AS CONDITIONED STABILITY

The framework separates two explanatory targets: (i) a STATE THEORY specifying what it is for an utterance type to be grammatical *for a population in a communicative situation* at time t ; (ii) a DYNAMICS MODULE specifying how such states tend to arise and persist. The state theory is constitutive; the dynamics module is etiological. Table 3 summarizes the notation.

Table 3: Notation summary

Symbol	Interpretation
<i>State theory</i>	
u	Utterance type or candidate token
$g \in \mathcal{H}$	Construction node in a hierarchy
$c \in \mathcal{C}$	Conditioning state (situation + norm-centre)
$\text{map}(u, c)$	Structural viability ($\in \{0, 1\}$)
$K(u, c)$	Operator-value coherence with reject option ($\in [0, 1]$)
$C_t(u, c)$	Situational licensing at time t ($\in [0, 1]$)
$\tilde{G}_t(u, c)$	Graded stability score ($= \text{map} \cdot K \cdot C_t$)
<i>Feeling model</i>	
$F_{i,t}(u, c)$	Subjective feeling of anomaly
$L(u)$	Dependency locality cost
$R_i(u, c)$	Processing cost vector (interference, locality, surprisal)
<i>Dynamics (§4)</i>	
$N_t(n, c)$	Number of opportunities for niche n in state c
$\rho_t^*(v \mid n, c)$	Counterfactual choice probability among analogical candidates
s_t, e_t, p_t	Positive evidence, error evidence, preemption mass

3.1 CONDITIONING STRUCTURE

Let $c \in \mathcal{C}$ be a CONDITIONING STATE – a construed communicative situation (Wiese (2023)) together with whatever norm-centre is treated as relevant. Nothing here requires c to be externally given; interlocutors can misalign about c , and c can be learned, split, and renegotiated over time.

For many purposes it is useful to think of c as a bundle that may include situational features (activity type, medium, footing), stable baselines tied to speaker ascription, and norm-orientation (discourse-community identification), but the formalism treats c abstractly as the conditioning variable that selects the relevant distribution of form–value relations.

3.2 OBJECTS

Let u range over utterance tokens or fine-grained utterance types, with structural representation $M(u)$. Let $\sigma(u, c)$ be the utterance-level interpretation made available in c (lexical content plus pragmatic enrichment, indexical resolution, discourse update, etc.).

Utterance types inherit licensing from a hierarchy of constructional nodes $\mathcal{H}$. A novel token such as *She texted him the address* need not have been encountered before; it can still be covered by licensed nodes for ditransitive transfer, pronominal object order, and the relevant lexical-class relations. Let $A(u) \subset \mathcal{H}$ be the nodes activated by u , and let $\lambda_g(u, c)$ be non-negative coverage weights with $\sum_{g \in A(u)} \lambda_g(u, c) = 1$. Each node g has its own licensing posterior. A token-level licensing estimate can then be derived by partial pooling:

$$C_t(u, c) = \sum_{g \in A(u)} \lambda_g(u, c) C_t(g, c).$$

The sum is schematic; a similarity kernel over representations would do the same work. What matters is that the learner updates constructional regions, not only memorized sentence types. This prevents the dynamics from predicting that every first-heard sentence has zero licensing.

The framework distinguishes three state quantities.

(I) **MAPPING VIABILITY.** Let $\text{map}(u, c) \in \{0, 1\}$ indicate whether structural analysis can yield at least one well-typed representation for u (category assignment + structural composition) throughout the derivation.

$$\text{map}(u, c) = \begin{cases} 1 & \text{if } \exists \text{ valid analysis } M(u) \text{ yielding } \mu(M) \\ 0 & \text{otherwise.} \end{cases}$$

map is explicitly pre-grammatical: it assigns near-zero weight to inputs that defeat structural analysis (e.g. word salad, category errors) but doesn't enforce situational licensing. Situational non-licensing belongs only in C_t .

(II) **COHERENCE AS SATISFIED CONCENTRATION OF INTERPRETATION.** Even when $\text{map}(u, c) = 1$, interpretation can be unstable: multiple incompatible construals may compete, or the best available construal may violate an operator constraint. A normalized distribution over ordinary interpretations would measure ambiguity but not violation: a single catastrophic construal would still receive probability 1. To avoid that bug, the model adds a null outcome ω_0 representing rejection or failed operator-value satisfaction. Let $\Omega^+(u, c) = \Omega(u, c) \cup \{\omega_0\}$ and fix $E(\omega_0; u, c) = \kappa_c$:

$$p(\omega \mid u, c) = \frac{e^{-E(\omega; u, c)}}{e^{-\kappa_c} + \sum_{\omega' \in \Omega(u, c)} e^{-E(\omega'; u, c)}}.$$

$$K(u, c) = \max_{\omega \in \Omega(u, c)} p(\omega \mid u, c) \in [0, 1].$$

I choose the null-outcome repair because it preserves the probabilistic interpretation of K . An unnormalized score $e^{-E(\omega^*)}$ or an explicit $K_{\text{peak}} \times K_{\text{satisfaction}}$ decomposition would make similar predictions, but the null outcome keeps ambiguity and satisfaction in a single race. A good single construal has $E(\omega^*) \ll \kappa_c$ and $K \approx 1$; a transparent but operator-violating construal has $E(\omega^*) \gg \kappa_c$, so probability mass shifts to ω_0 and $K \approx 0$.

The energy model $E(\omega; u, c)$ is *open-ended*:

$$E(\omega; u, c) = \sum_{j \in \mathcal{J}(c)} w_j \phi_j(\omega; u, c),$$

where $\mathcal{J}(c)$ includes constraints on grammaticalized features live in c . In the demonstrations below, I assume a minimal inventory: (1) temporal alignment where tense–aspect is grammaticalized, (2) argument-structure satisfaction where a construction licenses roles, (3) information-structural fit where topic/focus packaging is grammatically specified, and (4) indexical consistency only when the contrast belongs to a closed accountable repertoire. This criterion makes K partly derivative of licensing facts. A value can enter E only after the community has conventionalized it as a grammaticalized constraint.

This decomposition bridges the narrative diagnostic categories of §2.1 to the formal variables. For instance, coherence demands may be grouped by their subsystem – denoting the sum over purely semantic demands as E_{sem} and over indexical/pragmatic demands as E_{index} – to yield an expositional factorization $K \approx K_{\text{sem}} \cdot K_{\text{index}}$. In practice, however, the framework assumes that stability depends on the joint concentration of all constraints live in the communicative situation c .

(III) SITUATIONAL LICENSING (ENTRENCHMENT) UNDER CONDITIONING. For individual i at time t , let $\Lambda_{i,t}(u, c) \in \{0, 1\}$ indicate whether i *licenses* u as a community resource in c (i.e. treats the relationship as a legitimate option rather than a performance slip or nonce error).

Define the population-level licensing rate:

$$C_t(u, c) = \Pr_i(\Lambda_{i,t}(u, c) = 1) \in [0, 1].$$

This is the object previously called “entrenchment”. Here it is explicitly conditioned on c . The population quantified over by C_t isn’t a separate primitive; it is induced by the norm-centre and communicative situation bundled in c .

3.3 OBJECTIVE GRAMMATICALITY AS A CONDITIONED STATE PROPERTY

I write $\mathbb{I}[\cdot]$ for the indicator function: $\mathbb{I}[\phi] = 1$ if ϕ is true and 0 otherwise.

The state theory distinguishes a graded stability score from a categorical membership predicate.

3.3.1 A graded stability score

Recall that $\text{map}(u, c) \in \{0, 1\}$ indicates whether u has at least one available structural analysis in conditioning state c that yields a well-defined operator value. Let $K(u, c) \in [0, 1]$ be coherence as satisfied concentration of the interpretation distribution, and let $C_t(u, c) \in [0, 1]$ be the population licensing rate at time t in c .

Define the graded stability score:

$$\tilde{G}_t(u, c) = \text{map}(u, c) \cdot C_t(u, c) \cdot K(u, c) \in [0, 1].$$

This quantity is the workhorse for modeling gradience and for linking the state theory to behavioural measures.

3.3.2 Projectible use of the graded score

$\tilde{G}_t$ is not bookkeeping after the real explanation has happened. It is the macro-state over which grammaticality projects. Low- $\tilde{G}$ forms should be repaired, avoided, transmitted less faithfully, and routed into change trajectories regardless of whether the low score comes from mapping failure, operator-value incoherence, or non-licensing. High- $\tilde{G}$ forms should show the converse profile.

The feedback loop is simple. Production and repair behaviour at time $t + 1$ depend partly on $\tilde{G}_t$:

$$\begin{aligned} \Pr(\text{produce } u \mid n, c, t) &\propto \tilde{G}_t(u, c) \rho_t^*(u \mid n, c), \\ \Pr(\text{repair } u \mid c, t) &= r_0 + r_1(1 - \tilde{G}_t(u, c)), \quad r_1 > 0. \end{aligned}$$

Those behaviours generate the later evidence streams s_t , e_t , and p_t . This closes the micro-macro loop: individual production and repair update population licensing, and the resulting population state projects back into future production and repair. The projectible generalizations are therefore satiation profiles, transmission fidelity, repair rates, and change trajectories over $\tilde{G}_t$ as such, not only over its separate factors.

3.3.3 A categorical membership predicate

Communities often treat grammaticality as a constitutive membership fact: either a relationship counts as an available resource in c or it doesn't. A categorical predicate can be defined by thresholding the graded score:

$$G_t(u, c) = \mathbb{I}[\tilde{G}_t(u, c) \geq \tau(c)].$$

The threshold $\tau(c)$ isn't a hidden grammatical parameter. It is a decision criterion associated with the communicative situation: some situations are strict about what counts as “in the repertoire”, others are permissive.

3.3.4 A decision-theoretic motivation

Suppose a judge in c is choosing between two labels, $\ell \in \{\text{IN-REPERTOIRE}, \text{NOT-IN-REPERTOIRE}\}$, and faces asymmetric losses. Let $L_{\text{FA}}(c)$ be the loss of treating an item as IN-REPERTOIRE when it isn't, and $L_{\text{FR}}(c)$ the loss of treating an item as NOT-IN-REPERTOIRE when it is. Under the simplifying assumption that $\tilde{G}_t(u, c)$ is a calibrated signal of stability, the optimal rule is to accept u as IN-REPERTOIRE whenever:

$$\tilde{G}_t(u, c) \geq \tau(c) = \frac{L_{\text{FA}}(c)}{L_{\text{FA}}(c) + L_{\text{FR}}(c)}.$$

On this view, $\tau(c)$ is induced by payoff structure: high-stakes institutional contexts raise L_{FA} (and thus τ), whereas in-group, low-stakes contexts raise L_{FR} (lowering τ).

3.4 WHAT map DOES AND DOESN'T DO

The mapping predicate $\text{map}(u, c)$ is reserved for genuine analyzability failure: cases where no structural analysis is available that yields any well-defined operator value in c . This is the only categorical failure mode in the representational sense.

Importantly, phenomena traditionally described as categorical “structural bans” aren't treated here as mapping failures. In those cases, the intended analysis is typically available as a candidate

representation – so $\text{map}(u, c) = 1$ – and the intended construal can be coherent once stipulated (so $K(u, c)$ need not be small). What makes the pattern behave categorically in the community is instead the licensing term: under the relevant conditioning states, the community converges on near-universal non-licensing, $C_t(u, c) \approx 0$. This is the formal counterpart of what earlier drafts (and LingbuzzPreprint.tex) call “near-zero entrenchment”.

3.5 INTERLOCUTOR MISALIGNMENT ABOUT CONDITIONING

Because c is construed, interlocutors can disagree about which conditioning state is in force. Let a hearer h maintain a posterior $q_h(c | e)$ over conditioning states given cues e (situational cues, ascription cues, stance cues, etc.). Then the hearer’s expected stability score is

$$\tilde{G}_t^{(h)}(u | e) = \sum_{c \in \mathcal{C}} \tilde{G}_t(u, c) q_h(c | e).$$

3.6 IDENTIFYING THE CONDITIONING STATE BEFORE PREDICTION

The conditioning state c is not a free parameter to be selected after a judgment is observed. For empirical work, c has to be fixed before predicting ratings or repair behaviour. The posterior $q_h(c | e)$ is the identification device: it is estimated from non-judgment cues such as activity type, medium, participant roles, speaker ascription, register cues, and overt norm orientation. A form like *might could* is therefore not rescued by choosing a sympathetic norm-centre after the fact; the model predicts different judgments only when the independent cues make that norm-centre probable.

This also blocks a competence/performance redux. C_t is identified by corpus-rate-per-opportunity, elicited production, and repair behaviour, not by acceptability ratings alone. F is identified by ratings, online measures, and self-reports. The model is falsified where these dissociate in the wrong direction: if satiation changes later corpus rates for a purportedly stable gap, if repair behaviour tracks prescriptive dissonance rather than licensing, or if high-opportunity preempted forms behave like low-evidence uncertainties, the state assignment is wrong.

3.7 THE FEELING OF (UN)GRAMMATICALITY

Let $F_{i,t}(u, c) \in [-1, 0]$ be the subjective anomaly signal for individual i . Model it as a response to low stability plus implementation costs and ideological overlays:

$$F_{i,t}(u, c) = -\alpha(1 - \tilde{G}_t^{(h)}(u | e)) - \gamma^\top R_i(u, c) - \delta D_i(u, c) + \varepsilon_i,$$

where $h = i$ is the hearer processing the signal, and $\tilde{G}_t^{(h)}$ is the hearer’s expected stability score accounting for situational uncertainty (§3.5). $R_i(u, c)$ is a vector of processing/repair costs (locality, interference, garden-path reanalysis, surprisal, etc.). The locality component $L(u)$ sums over dependencies d_i :

$$L(u) = \sum_i \ell(|d_i|), \quad \ell(k) = \begin{cases} k, & k \leq K_{\text{sat}}, \\ K_{\text{sat}} + \beta(k - K_{\text{sat}})^\eta, & k > K_{\text{sat}}. \end{cases}$$

Here $|d_i|$ is the length of dependency i (in intervening discourse referents or words), K_{sat} is a saturation threshold beyond which additional distance contributes sublinearly, and β, η (with $0 < \eta < 1$) control the rate of that sublinear growth. $D_i(u, c)$ is prescriptive dissonance.

INTERPRETATION. $F = 0$ corresponds to a null affect (no anomaly reported), while $F < 0$ is a metacognitive warning proportional to the summed evidence of misfiring. The same feeling can arise from different mixtures of causes (structural, procedural, or social).

MEASUREMENT STRATEGY. F can be measured via explicit ratings (mapped linearly to $[-1, 0]$) or implicit probes like N400/P600 magnitudes and self-paced reading slowdowns (proxies for components of R_i).

3.8 MEASUREMENT MODEL FOR C_t

$C_t(u, c)$ is latent and is estimated from converging indicators rather than from ratings. Let $y_t(u, c)$ be a vector of observable traces (corpus rate per opportunity, elicited production probability, repair probability, and recognition as a legitimate resource). Repair probability enters with the sign predicted by §3.3.2: low- C_t resources should invite repair when participants orient to the relevant norm-centre. A minimal measurement model is

$$y_t(u, c) = b + \Lambda \logit(C_t(u, c)) + \epsilon.$$

3.9 WORKED EXAMPLES: DERIVING STATUS FROM COMPONENTS

The status of the four canonical examples (§2.2) can now be defined by computing their component values.

1. **Can the have running* (Nonsense): $\text{map} = 0$ because structural analysis fails to assign a valid category structure. Result: $\tilde{G} = 0$. Status: Ungrammatical.
2. **I've finished it yesterday* (Semantic Clash): $\text{map} = 1$ (well-formed syntax), $C_t \approx 1$ (components entrenched), but $K \approx 0$ due to the clash between Present Perfect and specific past adverbial (temporal alignment ϕ_{temp} fails). Result: $\tilde{G} \approx 0$. Status: Ungrammatical.
3. *?friend of whose* (Situationally-Novel): $\text{map} = 1$, $K \approx 1$ (perfectly interpretable), but C_t is low/indeterminate because the opportunity set N_t is small (low preemption mass p_t). The Beta update hasn't driven C_t to 0, leaving it in a state of "not established". Result: $\tilde{G}$ intermediate. Status: Marginal/Undefined.
4. **Which did you buy car?* (Stable Gap): $\text{map} = 1$, $K \approx 1$, but $C_t \rightarrow 0$ with high certainty. Since N_t is large (questions are frequent) and the counterfactual choice probability ρ_t^* is high, the effective preemption mass p_t is large. This drives the Beta posterior mass rapidly to zero (Figure 1). Result: $\tilde{G} \approx 0$. Status: Categorically Ungrammatical.

4 DYNAMICS: WHY THE STATE TENDS TO BECOME WHAT IT IS

The dynamics module explains trajectories of $C_t(u, c)$ (and, in principle, refinements of the conditioning partition itself). It doesn't define grammaticality.

4.1 NICHES, COMPETITORS, AND OPPORTUNITY SETS

Let n index a CONSTRUCTIONAL NICHE (a communicative job), and let $\mathcal{V}_{n,t}^*$ be the opportunity set: the observed, licensed, and analogically generated variants that could plausibly do that job in some conditioning states. Let $N_t(n, c)$ be the number of opportunities for niche n in conditioning state c over some time window, and $k_t(v, n, c)$ the observed count of variant $v \in \mathcal{V}_{n,t}^*$.

The star matters. Stable gaps and innovations often involve variants with no positive tokens. Their counterfactual utilities are still defined because the hierarchy $\mathcal{H}$ generates analogical candidates. English learners can represent a left-branch extraction candidate by combining wh-fronting,

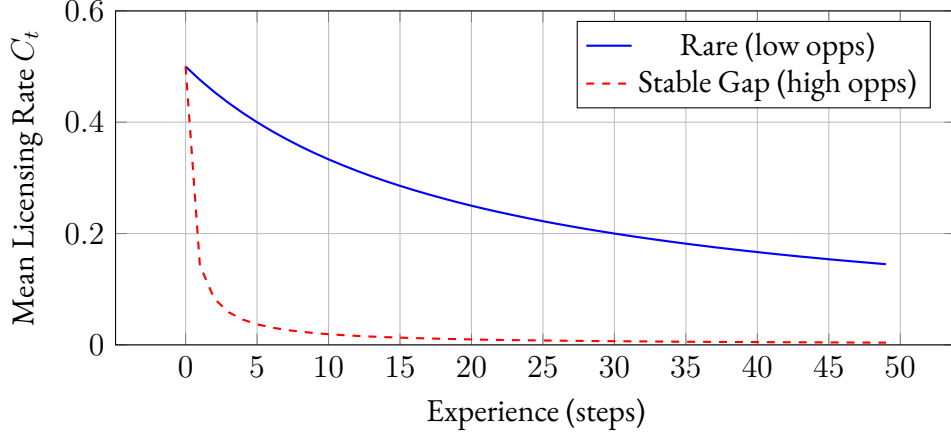


Figure 1: Deterministic posterior means for situational licensing rate C_t under preemption. When effective preemption mass is high (Stable Gap, red), C_t is driven rapidly to zero. When opportunities are rare (Situationally-Novel, blue), C_t declines slowly, leaving the construction in a state of indeterminacy rather than categorical rejection.

modifier–noun packaging, and question formation, even if no one licenses that candidate as an English resource. The same machinery lets a novel but ordinary sentence inherit coverage from licensed higher-grain constructions rather than being preempted merely because that exact sentence type is new.

4.2 USAGE AS LICENSING $\times$ CHOICE AMONG CANDIDATES

Separate LICENSING from SELECTION. A variant can be licensed but rarely chosen.

Let $\rho_t^*(v \mid n, c) \in [0, 1]$ be the counterfactual probability of choosing v if it were available in niche n under c . A flexible choice model is a softmax over utilities:

$$\rho_t^*(v \mid n, c) = \frac{e^{U_t(v; n, c)}}{\sum_{v' \in \mathcal{V}_{n, t}^*} e^{U_t(v'; n, c)}}, \quad U_t(v; n, c) = \boldsymbol{\theta}^\top \mathbf{f}(v; n, c).$$

At the population level, the expected usage rate $\pi_t(v \mid n, c) \approx C_t(v, c) \cdot \rho_t^*(v \mid n, c)$. The counterfactual choice term can be high even when $C_t(v, c)$ is near zero; that is exactly the condition under which absence becomes informative.

4.3 DYNAMICS: PREEMPTION AS EFFECTIVE PREEMPTION MASS

The core idea is that learners update licensing not only from positive uses of u but also from structured non-occurrence when u is a plausible competitor and is repeatedly not chosen. The update shouldn't treat all "opportunities" as equally informative: absence is evidential only to the extent that u would have had non-trivial probability of being chosen if it were licensed.

For a target variant u associated with niche $n(u)$, define the EFFECTIVE PREEMPTION MASS in a time window as the expected number of u -tokens that would have been produced if u were a licensed competitor:

$$p_t(u, c) = \sum_{j=1}^{N_t(n(u), c)} \rho_t^*(u \mid n(u), c),$$

where $N_t(n(u), c)$ is the number of niche opportunities observed in that window. This quantity is large only when (i) the opportunity set is large and (ii) u would have been chosen at non-trivial rates if licensed.

A simple Bayesian learner for licensing represents each construction node (g, c) with a Beta posterior $\text{Beta}(a_{g,t}, b_{g,t})$, updated by three evidence streams: s_t (positive evidence), e_t (error evidence), and p_t (effective preemption mass). An observed token or missed opportunity distributes its evidence over activated nodes by the coverage weights $\lambda_g(u, c)$:

$$\begin{aligned} a_{g,t+1} &= a_{g,t} + \lambda_g(u, c) s_t(u, c), \\ b_{g,t+1} &= b_{g,t} + \lambda_g(u, c) \{e_t(u, c) + p_t(u, c)\}. \end{aligned}$$

The implied node-level mean licensing rate is then:

$$C_{t+1}(g, c) = \frac{a_{g,t+1}}{a_{g,t+1} + b_{g,t+1}}. \quad (29)$$

Token-level $C_t(u, c)$ is obtained by the partial-pooling equation in §3.2.

The mean-field approximation. While I treat the discrete update (29) as the primary dynamics, it is sometimes useful to approximate the expected trajectory of the posterior mean $C_t(g, c)$ by a deterministic ODE. Under stationarity and large-sample assumptions, replacing the stochastic evidence streams by their expectations yields an approximation of the form $\dot{C} = r(g, c) C(1 - C)$, where $r(g, c)$ is a net evidence rate (roughly, expected positive evidence minus expected preemption/error evidence per unit time, with ρ_t^* determining the effective preemption mass). I use this ODE only for qualitative fixed-point and comparative-statics arguments.

4.4 EMERGENT CATEGORICALITY AS BIMODALITY

The sense of EMERGENCE used here is weak and mechanistic. Population-level categoricity is derived from learner-level updating, selection, and repair; it is not a label for whatever remains unexplained. This differs from Hopper’s “emergent grammar”, where grammar is perpetually provisional (Hopper, 1987). The present model allows conditioned states with near-absorbing attractors.

In the mean-field approximation, $\dot{C} = rC(1 - C)$ has fixed points at 0 and 1. When $r > 0$, positive evidence dominates and the licensing posterior is driven toward 1. When $r < 0$, error evidence and effective preemption mass dominate and the posterior is driven toward 0. Across construction nodes, a community with many evidence rates bounded away from zero therefore develops a bimodal distribution of C_t : most nodes cluster near licensed or non-licensed attractors, while only low-evidence or contested nodes remain diffuse.

This is the formal sense in which categorical grammaticality emerges. Closure raises ρ_t^* by keeping the competitor set small and role-specific; large $N_t(n, c)$ supplies repeated opportunities; their product is the effective preemption mass. Closed, high-opportunity, update-critical contrasts are therefore exactly the parameter regime in which the licensing posterior becomes bimodal. A finite-population stochastic version should strengthen this pattern: sampling drift near 0 and 1 makes the boundaries sticky, while middle states remain vulnerable to renewed evidence. This connects the present dynamics to complex-adaptive and cultural-evolutionary accounts of language change (Beckner et al., 2009; Blythe & Croft, 2012; Kirby et al., 2014).

4.5 HOW APPARENT CATEGORICAL GAPS ARISE WITHOUT HARD BANS

When $\text{map}(u, c) = 1$ and $K(u, c)$ is high but $C_t(u, c)$ is driven toward zero by persistent preemption in a large opportunity set, speakers converge on a sharp, non-satiating rejection profile. Additional processing penalties in $R_i(u, c)$ (e.g.

systematic garden-path repair due to an entrenched fused-head construal) can strengthen the subjective categoricity without any independent structural veto being posited.

This approach transforms the framework from a descriptive theory into a quantitative model capable of precise predictions about grammaticality judgments and their evolution over time.

5 THEORETICAL IMPLICATIONS

MVMG yields several important implications for the understanding of grammaticality:

First, it supports treating grammaticality as an emergent property in the formal sense of §4.4: discrete-looking judgments arise when licensing posteriors over constructional nodes become bimodal. Different patterns in the distribution of grammatical constructions across adjective types illustrate the same point. The distribution of *much* versus *more* with different adjective classes (comparative governors, participial adjectives, etc.) follows real patterns, but those patterns arise from interactions among modifier semantics, adjective scale structure, and discourse-pragmatic factors (Reynolds, 2024). This helps explain why efforts to reduce grammaticality to simple necessary and sufficient conditions have repeatedly fallen short.

Second, this framework clarifies how formal and usage-based approaches capture different aspects of grammatical stability. The stability of grammatical patterns depends both on their internal systematic properties (emphasized by formal approaches) and on their role in meeting communicative needs (emphasized by functional accounts). Many grammatical phenomena exhibit stability patterns that can't be reduced to local collections of features. Discourse context can change whether a construction maintains stable form–value relationships, because the relevant operator values are conditioned by the activity, norm-centre, and public-update role. This helps explain why purely local syntactic models often fail to capture the full range of grammaticality phenomena.

Third, the analysis suggests specific predictions about how grammatical stability is maintained and lost. If grammatical constructions are maintained through multiple interacting factors, one should expect:

1. Instability to manifest in coordinated ways across multiple properties rather than through isolated changes
2. Periods of gradually increasing instability followed by relatively rapid reorganization when stability thresholds are crossed
3. Different but equally stable grammatical patterns arising in different language communities
4. Gradient effects in grammaticality judgments reflecting varying degrees of stability

Fourth, MVMG illuminates the relationship between competence and performance. Rather than treating these as fundamentally different phenomena, they can be understood as different manifestations of the same stability conditions. Processing limitations and other performance factors help shape which form–value relations become stable, while those stable relationships in turn constrain possible performance patterns. The identification strategy in §3.6 keeps this from becoming an escape hatch: licensing and feeling are measured by different evidence streams and can falsify each other.

Finally, this approach offers a new perspective on systematic constraints in grammar, such as the

English ban on left-branch extraction. Instead of viewing these as either innate rules or processing limitations, they can be understood as particularly robust stability conditions in form–value relations. Their persistence reflects deep patterns of stability, while cross-linguistic variation shows how different stable solutions can emerge in different communities.

These implications suggest concrete directions for future research. More detailed studies are needed of how multiple properties interact to create and maintain stable form–value relations, better methods for measuring degrees of stability, and closer examination of the transition points where grammatical systems reorganize. The framework also calls for renewed attention to variation across language communities, as different stability patterns may shed light on the fundamental nature of grammatical organization.

The crucial idea is that grammaticality represents a real linguistic kind: a projectible profile of conditioned form–value stability maintained through usage within communicative situations. It earns its keep by supporting predictions about repair, transmission, satiation, and change trajectories over $\tilde{G}_t$ as such (§3.3.2), not merely by naming a cluster of mechanisms.

5.1 ACTUATION AND THE DYNAMICS OF GRAMMATICAL CHANGE

The concept of *actuation* can now be defined as a sign change in the expected evidence balance for licensing. Informally, actuation occurs when the expected positive stream $s_t(u, c)$ begins to outweigh the combined negative streams $e_t(u, c) + p_t(u, c)$ across successive windows, so that the posterior mean $C_{t+1}(u, c)$ in (29) increases rather than decays. In the mean-field approximation (box in §4.3), this corresponds to the regime in which the net evidence rate $r(u, c)$ becomes positive, yielding the familiar S-curve at the population level.

The factors that drive such bifurcations align with the motivations discussed in §2.5. Semantic reanalysis may increase utility (and thus ρ_t^*) by making a form–value relation more transparent or useful. Social pressures may shift utilities (prestige effects) or alter the relevant community standard. Structural motivations such as analogical extension can systematically raise licensing probability by borrowing strength from frequent neighbors. Processing innovations may reduce error rates (e_t) or locality costs, improving the net evidence balance.

Actuation isn’t simply a matter of individual innovation but requires coordinated community-level change. A few speakers adopting a marginal construction doesn’t guarantee its success; actuation demands that the underlying dynamics shift such that acceptance systematically outweighs rejection across the speech community. This explains why many innovations fail despite being individually sensible – they appear before the community conditions are right for the evidence balance to become positive.

The bifurcation framework also clarifies why certain changes appear to accelerate once they begin. Near the critical point where evidence streams balance, small perturbations in community attitudes can trigger rapid shifts between rejection and acceptance. As the evidence balance becomes more strongly positive, the construction moves further from the unstable rejection state, making reversal increasingly unlikely. This creates the characteristic S-curve pattern observed in many documented language changes (Kroch, 1989), derived from the population dynamics in §4.4.

For constructions currently existing in the marginal zone – those with near-zero evidence balance – the framework predicts heightened sensitivity to external factors and greater cross-community variation. Small speech communities may show particularly volatile behaviour near bifurcation points, as stochastic effects can overwhelm weak deterministic trends. This may explain why certain changes

appear to originate in geographically or socially peripheral communities before spreading to larger population centres.

5.2 MACRO-TYPOLOGICAL CONSTRAINTS ON GRAMMATICAL DESIGN

Recent macro-typological work sharpens the question of how strongly grammar is constrained across languages. Verkerk et al. (2025) test 191 implicational universals from the Universals Archive against Grambank’s 2,430-language morphosyntactic sample, using Bayesian models that control explicitly for genealogical and areal non-independence and then follow up with spatiophylogenetic analyses of evolutionary rates. A naïve analysis that ignores relatedness appears to support the vast majority of proposed universals, but once phylogeny and geography are accounted for, only about a third (60 of 191) remain statistically supported. Support is concentrated in relatively narrow domains: most hierarchical universals and a substantial minority of “narrow” word-order universals survive, whereas “broad” word-order universals and miscellaneous others fare poorly. Diachronically, supported universals typically correspond to “harmonic” combinations of features (for instance, consistent head-dependent orders) that languages are more likely to evolve into than out of, with these preferred configurations recurring independently across lineages.

From an MVMG perspective, these findings identify recurrent attractors in the global design space of operator-valued form–value relations rather than exceptionless grammatical laws. On the present account, such attractors correspond to configurations for which the net evidence balance (positive evidence s_t vs. preemption/error mass $e_t + p_t$) in the dynamics of §4.3 is positive across a wide range of communities. They are cognitively and communicatively favourable enough that repeated episodes of change tend to push $C_t(u, c)$ toward fixation in lineage after lineage. The fact that roughly two-thirds of the tested universals fail once autocorrelation is controlled for also fits a de-idealized view of grammaticality. Community-specific conventions still have considerable freedom in how they realize operator values, with only some regions of the space strongly preferred. Large-scale comparative work of this kind therefore complements MVMG: it circumscribes which parts of the operator-value design space are globally stable, while the present framework provides a meso-level account of how local community dynamics and form–value coherence give rise to those long-run macro-typological regularities.

5.3 RELATIONSHIP TO GENERATIVE GRAMMAR

The generative tradition has provided linguistics with foundational contributions into the systematic nature of grammatical knowledge. Most fundamentally, it demonstrates that grammaticality can’t be reduced to semantic plausibility, processing ease, or frequency of attestation. When Chomsky introduced *Colorless green ideas sleep furiously*, he showed definitively that speakers can recognize syntactically well-formed sentences even when they are semantically bizarre. The framework coherently explains categorical constraints like the impossibility of extracting determiner-adjective sequences in English (**Which do you prefer car?*), and critically, it captures the fact that such sequences remain ungrammatical even when their intended meaning is clear and processing demands are low.

MVMG shares with generative grammar several foundational observations: that grammaticality judgments reflect systematic, real patterns (Dennett, 1991); that these patterns can’t be reduced to meaning or processing alone; and that certain syntactic configurations appear to be categorically excluded regardless of context. The analysis of multiple center embeddings presented in §2.3.1 builds directly on ideas about recursive structure, and my treatment of stable gaps (§2.3.1) acknowledges

the generative discovery that some constructions appear to be universally excluded by the grammar itself.

The generative tradition has also identified fascinating puzzles that any theory of grammar needs to address, such as the independent relative *whose* (1d). As Hankamer and Postal (1973) observe, this construction appears to violate no syntactic principles: independent genitives are possible (*Mine was visiting*), independent interrogative *whose* is grammatical (*Whose was open?*), and *whose* functions perfectly well as a dependent relative pronoun (*the student whose friend was visiting*). The generative tradition's careful documentation of such cases, where seemingly parallel constructions show puzzlingly different grammatical status, has been invaluable in pushing theoretical development forward.

Where MVMG departs from generative grammar is in its explanation of such patterns. Rather than positing an autonomous syntactic component, MVMG suggests that grammatical constraints are produced by the interaction of form–value relations within specific communicative situations. The independent relative *whose* construction illustrates this difference. For this construction to be felicitous, multiple conditions have to converge: the possessor has to be sufficiently accessible in the discourse while the possessum is predictable enough to license ellipsis, but the possessive relationship needs to be semantically significant enough to warrant explicit marking, and this configuration has to occur in a context where a relative clause is the optimal way to package this information. The extreme rarity of contexts satisfying all these conditions appears to prevent the construction from becoming conventionalized in the grammar at all – speakers encounter it so rarely, despite perfectly common components, that even when all conditions align perfectly, the construction feels alien.

This approach draws on the generative observations about systematic constraints while providing a different perspective on their source. Where generative theory has to explain why a syntactically possible and pragmatically useful construction is systematically avoided, MVMG suggests that extreme mismatches between predicted and observed frequency may themselves be evidence of strong negative weighting, even when the exact source of that evidence remains unclear. The framework thus preserves what is most valuable in generative theory – its recognition of systematic grammatical constraints – while embedding those insights in a broader theory of how form–value relations become established and maintained in language communities.

Other cases that generative grammar struggles to explain are those that are grammatical with one meaning but ungrammatical with another, such as (13) *I have 16 years*. While syntactically identical to grammatical expressions like *I have 16 dollars*, this construction becomes ungrammatical specifically when used to express age. A purely syntactic account would have to explain why the same structure is well-formed in one case but ill-formed in another, despite no apparent syntactic differences. MVMG, in contrast, locates the source of ungrammaticality in the community's form–value conventions: *have*+numeral years has become conventionalized for expressing duration or future time (*I have 16 years until retirement*) but lacks positive evidence for expressing age, where a different construction (*I am 16 years old*) is the established pattern. Similar cases arise with plural forms that are grammatical with some meanings but not others (e.g. *peoples* for ethnic groups but not multiple individuals) and with verbs that resist certain arguments despite no obvious syntactic prohibition (e.g. *discuss about*). These meaning-dependent grammaticality patterns suggest that what gets down-weighted or up-weighted often depends on specific form–value associations rather than purely structural constraints.

5.4 RELATIONSHIP TO CONSTRUCTION GRAMMAR

Construction Grammar (CxG) (Fillmore, 1988; Fillmore et al., 1988; Goldberg, 1995, 2019; Kay & Fillmore, 1999; Sag, 2012) represents a significant theoretical advance in the understanding of linguistic knowledge. At its core, CxG argues that language consists of learned relationships between form and meaning at multiple levels of complexity. These form–value relations, or constructions, range from individual morphemes to abstract syntactic patterns. This perspective helps explain phenomena that proved challenging for earlier approaches, including idiomatic and partially schematic expressions that don’t fit a clean core/periphery split.

CxG’s most valuable contribution lies in demonstrating that meaning suffuses all levels of grammatical organization. Rather than treating syntax as an autonomous formal system that interfaces with semantics only at designated points, CxG reveals how meaning and form are inseparable aspects of linguistic knowledge. For instance, the *What’s X doing Y?* construction carries an implication of incongruity that can’t be derived from its component parts (Kay & Fillmore, 1999). Recent work connects this framework to model-internal analysis: Weissweiler et al. (2023) use construction grammar to probe how neural language models handle different levels of linguistic abstraction.

MVMG shares these CxG commitments about form–value relations and constructional abstraction. Its difference is not that one substance class is intrinsically privileged. The companion operator-stratum paper explicitly denies a substance-based boundary: phonological, gestural, lexical, morphological, and syntactic material can all realize operator contrasts when they enter a closed, accountable public-update repertoire (Reynolds, 2026). The MVMG claim is narrower. Categorical grammaticality clusters around form–value relations with a particular role: they are high-opportunity, closed-paradigm contrasts that configure update, role allocation, scope, reference, or repair.

This difference becomes clear when speakers judge violations. Register, politeness, genre, and many lexical choices can be stable and socially important without blocking the public update itself. A wrong tense, agreement value, clause-linking marker, or obligatory allomorph can mis-set the update instruction even when the intended message is recoverable. The age-predication example (*I have 25 years*) is transparent but not licensed for that English frame; its rejection comes from non-licensing in a high-opportunity constructional niche, not from the substance of syntax or morphology as such.

MVMG therefore preserves CxG’s systematic treatment of constructions while adding a prediction about which constructional contrasts attract categorical policing. The relevant contrasts are those whose closure and opportunity structure drive the licensing posterior toward the bimodal regimes in §4.4. That lets CxG’s continuum of constructions coexist with a non-arbitrary explanation of why some violations are heard as grammar and others as style, stance, or appropriateness.

5.5 RELATIONSHIP TO USAGE-BASED APPROACHES

MVMG shares with usage-based approaches (UBA) (J. Bybee, 2006, 2010; J. L. Bybee, 2007) the idea that linguistic knowledge develops from patterns of actual language use rather than from an autonomous formal system. Both perspectives reject the notion that grammaticality can be reduced to abstract rules operating independently of meaning and context. MVMG adds two distinctions that usage-based work often leaves implicit: licensing versus selection, and raw frequency versus frequency per opportunity.

The analysis of independent relative *whose*, as in *‘I saw Joan, a friend of whose was visiting* is a case in point. A simple UBA account might predict that this construction’s marginality stems from its low frequency. But this explanation proves insufficient: the construction isn’t merely rare but

dramatically rarer than one would expect given the frequency of its component parts. Independent *whose* appears in interrogatives (*Whose is that?*), and the relative *whose* is common in dependent contexts (*the student whose paper was late*). Given these frequencies, analogical extension should make the independent relative use more common than it is. The extreme rarity of independent relative *whose* in corpora, despite the grammatical availability of comparable elements and contexts, marks it as more than just infrequent – it points to a systematic gap in form–value relations.

The difference is not that MVMG exempts the operator stratum from usage dynamics. It makes the dynamics more specific. A rare form can remain grammatical when the opportunity set is tiny or when higher-grain constructions license the token by partial pooling. A transparent candidate can become categorically rejected when the opportunity set is large, the analogical candidate would have had high ρ_t^* , and an entrenched competitor wins every time.

This position lets MVMG incorporate usage-based findings about frequency and entrenchment while predicting when frequency becomes preemption. Cases like independent relative *whose* demonstrate the need for a theory that distinguishes simple rarity from systematic non-licensing. The same machinery explains why high-opportunity operator contrasts often yield sharp judgments: the evidence stream is dense enough for effective preemption mass to accumulate.

5.6 RELATIONSHIP TO LOGICALITY OF LANGUAGE ACCOUNTS

An alternative perspective, prominent in recent formal semantics, suggests certain types of unacceptability stem from the language faculty itself possessing a deductive system that identifies and filters sentences with logically trivial meanings (tautologies or contradictions) (cf. Chierchia, 2013; Del Pinal, 2019; Fox, 2000). This “logicality of language” hypothesis attempts to explain, for example, systematic restrictions on quantifiers by arguing the unacceptable cases are “L-trivial” – their triviality arises solely from the meaning and configuration of logical/functional terms (like *every*, *some*, *not*), irrespective of the open-class words (like *student*, *run*) (Gajewski, 2009). A key challenge is explaining why simple tautologies or contradictions (e.g. *It is raining and it is not raining*) are often acceptable. Del Pinal (2019) argues against “Logical Skeletons” (which assume the system ignores open-class word identity) in favour of “LF+RESCALE”, a view where the system sees standard logical forms but allows optional, context-dependent modulation of open-class terms (e.g. interpreting the second ‘raining’ as ‘raining hard’) to yield non-trivial meanings.

MVMG offers a potentially broader and more unified account. While ‘logicality’ approaches excel at explaining restrictions tied to functional vocabulary via L-triviality, MVMG aims to cover a wider spectrum of ungrammaticality, including cases not easily reducible to logical contradiction, such as absent form–value relations (1a), strong deviations from conventional community forms (17), or extreme, unexpected rarity (1d). By grounding grammaticality in community-specific conventions (§2.1.1) and allowing for gradient compatibility ($K(u, c)$) and acceptance ($C_t(u, c)$), MVMG inherently accommodates cross-linguistic variation and degrees of acceptability, aspects less central to the L-triviality filter. While LF+RESCALE provides a specific mechanism for acceptable ‘trivialities’, MVMG suggests these might arise from more general principles of interpretation within community norms, potentially avoiding the need to posit a dedicated deductive module and specific operators like RESCALE, and offering a clearer distinction between objective grammatical status and subjective processing effects or ‘feelings’ (§3.7). Thus, MVMG frames grammaticality as an emergent consequence of communicative practice in the formal sense of §4.4, rather than a direct output of logical computation within syntax.

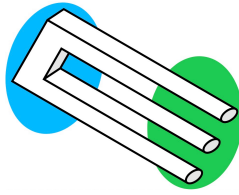


Figure 2: Impossible trident

5.7 RELATIONSHIP TO RELEVANCE-THEORETIC ACCOUNTS

Recent work by Scott-Phillips (2026) offers a fundamental observation: linguistic intuitions about acceptability arise as byproduct effects of cognitive systems for interpreting communicative acts. Just as people immediately sense when a visual stimulus violates core assumptions about physical objects (as with impossible objects), they detect when utterances violate basic presumptions about communicative efficiency. Significantly, Scott-Phillips argues that unacceptability occurs not from mere inefficiency, but from an inherent impossibility of interpreting an utterance consistently with these presumptions – similar to how an impossible trident (Figure 2) can't be interpreted as physically cohesive in any context.

MVMG shares several key premises with this account. Both frameworks reject the need for an innate grammar faculty, locating linguistic intuitions instead within general cognitive systems. Both recognize that language develops from communicative needs rather than autonomous syntactic principles. MVMG's emphasis on situation-specific form–value relations builds directly on Scott-Phillips's arguments about how communicative pressures shape linguistic conventions.

The frameworks differ primarily in their explanatory mechanisms. Where Scott-Phillips argues that grammaticality judgments reduce to impossibilities of efficient interpretation, MVMG suggests that while communicative pressures shape which form–value relations become conventionalized, these relationships then create systematic constraints that can't be reduced to efficiency alone. For Scott-Phillips, one would have to demonstrate that (1c) contains inherent contradictions making efficient interpretation impossible. MVMG instead analyzes how the present perfect's tense–aspect operator value clashes with the meaning of the lexeme *yesterday*. While these analyses might ultimately converge, it remains unclear why the criterion of inherent impossibility of interpretation should apply specifically to operator-value violations rather than to lexical-lexical conflicts or certain phonological patterns. The scope of what constitutes an interpretive impossibility requires further theoretical development.

This difference has important empirical implications. MVMG makes relatively concrete demands: it is possible to test whether specific form–value relations are stable within a communicative situation and identify precise points of operator-value conflict. The challenge for relevance-theoretic accounts, as Scott-Phillips (personal communication, Dec. 16, 2024) acknowledges, lies in establishing independent, empirically vulnerable claims about what makes efficient interpretation inherently impossible rather than merely difficult. Future work comparing specific predictions of each approach – particularly around how novel constructions become acceptable or unacceptable – could help clarify their relationship and complementary insights.

5.8 PREDICTIONS

The model’s distinctive predictions come from separating positive evidence, opportunity structure, and counterfactual choice probability. Cross-linguistic gender remains a useful boundary case, but it is not the strongest test. Any framework that treats grammaticalized concord as obligatory predicts stronger judgments where concord is more tightly grammaticalized. The companion operator-stratum account also complicates the simple gender story: gender concord can be policed through paradigm inertia even when its public-update value is attenuated (Reynolds, 2026). The sharper predictions concern stable gaps, artificial learning, and L2 transfer.

SATIATION FRAMING. Repeated exposure should attenuate negative feelings most strongly when a form is low-evidence rather than strongly preempted. Independent relative *whose*, as in *“The packages are still here, but Nathan, whose was open, just left.”*, should be sensitive to framing: participants told that tokens come from an intentional dialectal, literary, or ingroup resource should show more improvement than participants told that the same tokens are errors. Left-branch extraction should resist the same treatment because its opportunity set is large and its effective preemption mass is high. The dependent variable is not only rating change but later repair behaviour and elicited production.

ARTIFICIAL-LANGUAGE LEARNING. An artificial-language-learning experiment can hold positive evidence constant at zero while manipulating opportunity-set size. Learners could see a system in which a target variant is never produced. In the high-opportunity condition, competitor variants repeatedly occupy the relevant niche and the target has high ρ_t^* ; in the low-opportunity condition, the niche rarely arises or the target would have low counterfactual utility. MVMG predicts categorical rejection in the first condition and uncertainty in the second, even though both conditions contain identical zero positive evidence. This puts numbers on the Figure 1 contrast and is compatible with artificial-language learning paradigms that manipulate simplicity and communicative structure (Culbertson & Kirby, 2016).

L2 PREEMPTION TRANSFER. Early L2 rejections should track L1 preemption mass, not only L2 frequency. A learner whose L1 strongly preempts a candidate in a high-opportunity niche should reject the L2 analogue early, even if the L2 input frequency is low enough to leave native speakers uncertain. Conversely, low L1 preemption should make learners more willing to treat rare L2 patterns as unresolved rather than impossible. As learners join the L2 speech community, $q_h(c | e)$ and the constructional hierarchy should shift toward the target norm-centre. This view aligns with Selinker’s (1972) concept of INTERLANGUAGE, but gives a more specific prediction: transfer is strongest where the L1 has accumulated large effective preemption mass.

6 LIMITATIONS

While the proposed framework offers a comprehensive account of grammaticality, integrating insights from various theoretical traditions, it’s not without limitations. One potential limitation lies in the framework’s reliance on the concept of “communicative situation”. Defining the relevant conditioning context and determining the precise set of shared norms that govern grammaticality judgments can be challenging. As noted in §2.1.1, speakers navigate overlapping contexts that shift on multiple dimensions – immediate situation, durable social positions, and the norms they orient to. The framework acknowledges this fluidity and heterogeneity, but further research is needed to develop more precise methods for operationalizing these dimensions and measuring their influence on grammatical stability.

The framework, in its current form, may not fully capture the complexities of stylistic variation and individual preferences. While it accounts for broad patterns of acceptability and rejection, it doesn't delve deeply into the nuances of stylistic choices that fall within the realm of grammatical acceptability. Future refinements could incorporate a more fine-grained model of stylistic variation and its interaction with core grammatical principles.

Another potential limitation concerns the framework's ability to account for purely formal constraints that seem to exist independently of meaning or communicative function. While the paper argues that many seemingly arbitrary restrictions can be explained by historical processes and the interplay of various motivations, there may be residual cases of purely formal constraints that resist explanation within the current framework. Further investigation into these cases could lead to a more complete understanding of the factors shaping grammatical systems.

Despite these limitations, the proposed framework provides a robust and flexible foundation for understanding the multifaceted nature of grammaticality. It offers a promising avenue for future research, and the limitations outlined here serve as points of departure for further refinement and elaboration.

7 CONCLUSION

This paper proposes a novel framework for understanding grammaticality, one that moves beyond the traditional competence–performance dichotomy and embraces the dynamic interplay of form, meaning, processing constraints, and sociolinguistic factors. By conceptualizing grammaticality as an emergent property in the sense derived in §4.4, the framework accounts for both the categorical and gradient aspects of grammaticality judgments. It explains why some constructions are rigidly ungrammatical, others fluctuate between marginal and acceptable, and still others evolve into stable, conventionalized patterns.

The framework's core tenets – that grammaticality involves conventional form–value relations, that these relationships interact with processing constraints and sociolinguistic factors, and that different types of violations arise from different mismatches between form and meaning – provide a principled basis for understanding a wide range of linguistic phenomena. The analysis of examples drawn from both formal syntax and experimental data has demonstrated the framework's explanatory power, illuminating challenging cases that have resisted unified explanation in previous approaches.

The framework generates testable predictions about which ungrammatical constructions might change over time and offers potential applications for language teaching, clinical linguistics, and language documentation. It predicts that satiation effects should be more readily inducible for low-evidence constructions than for high-opportunity stable gaps, that artificial-language learners should distinguish zero evidence from strong preemption, and that early L2 rejections should track L1 preemption mass rather than L2 frequency alone.

Integrating the methodologies outlined in the previous section offers a path toward a more comprehensive understanding of how form–value relations become stable within communicative situations, what factors promote or inhibit their entrenchment, and how cross-linguistic variation arises. Future work can refine the metrics for identifying stable form–value mappings, develop computational models to predict the emergent regularities derived in §4.4, and expand the empirical base to underrepresented languages and language contact situations.

While acknowledging the limitations of the current formulation, this paper argues that the proposed framework represents a significant step toward a more comprehensive and nuanced understanding of grammaticality. By integrating insights from generative, functional, and usage-based approaches, it offers a unified perspective that recognizes the multifaceted nature of linguistic knowledge and its grounding in both individual cognition and usage within communicative situations.

Future research should focus on further refining the framework, developing more precise methods for measuring form–value stability, and investigating the complex interactions between different types of motivations (semantic, social, structural, and iconic). Cross-linguistic studies should prioritize contrasts whose operator status differs across languages, because the model predicts categoricity where closed, high-opportunity, update-critical repertoires form.

Ultimately, the study of grammaticality offers a window into the fundamental workings of human language. By de-idealizing grammaticality and embracing its dynamic, situation-relative nature, it becomes possible to gain a deeper understanding of the cognitive and social forces that shape language. This framework provides a robust foundation for such investigations, paving the way for a more integrated and comprehensive understanding of what it means for an utterance to be considered part of a language.

A TURKISH VOWEL HARMONY AND OPERATOR-EXPONENT REALIZATION

Turkish illustrates a sharp distinction between LEXICAL disharmony inside stems and ALLOMORPHIC harmony on inflectional suffixes.³ Only the latter interacts with grammaticality in the sense of the MVMG because suffixal harmony is part of how an inflectional operator exponent is realized.

STEM-INTERNAL VOWELS: DISHARMONY TOLERATED. Loanwords such as *doktor* ‘doctor’ violate backness harmony, but are fully acceptable; the language community simply memorises the form, so $C(u) = 1$ and $K(u) = 1$.

- (30) doktor
 doctor
 ‘doctor’ (disharmonic stem, grammatical)

Because no operator feature is left unrealized, the form–value mapping succeeds and

$$\tilde{G}_t(u, c) = C_t(u, c) \cdot K(u, c) \cdot \text{map}(u, c) = 1.$$

SUFFIXAL HARMONY: OPERATOR-EXPONENT REQUIREMENT. Inflectional morphemes are lexicalized with an underspecified vowel; the correct allomorph has to copy $[\pm\text{BACK}]$ (and, for some suffixes, $[\pm\text{ROUND}]$) from the final stem vowel. Using the “wrong” vowel leaves part of the exponent-selection requirement unsatisfied, so the null outcome in the K calculation wins and the word is ungrammatical.

- (31) a. kitap-lar
 book-PL.BACK
 ‘books’ (harmonic, grammatical)
 b. * kitap-ler
 book-PL.FRONT
 intended “books” (suffix harmony violation)
- (32) a. gül-dü
 laugh-PST.FRONT.ROUND
 ‘s/he laughed’ (harmonic, grammatical)
 b. * gül-du
 laugh-PST.BACK.ROUND
 intended “s/he laughed” (suffix harmony violation)

MVMG ACCOUNT. For the ill-formed **kitap-ler* and **gül-du*:

- *Mapping* succeeds: the plural / past node selects an exponent.
- *Compatibility* $K(u, c) \approx 0$: the chosen allomorph fails to realize the $[\text{BACK}]$ feature copied from the stem.
- *Community acceptance* $C_t(u, c) = 1$: every speaker knows how plural and past are expressed.

³See (Kabak & Vogel, 2001; Sezer, 1981) for discussion of the phonology and (Arik, 2015) for experimental evidence on native judgments.

Hence $G_t(u, c) = 0$ and speakers judge the word as categorically wrong, not merely odd-sounding. By contrast, *doktor* in (30) keeps $K(u, c) = 1$; harmony is a phonotactic preference that affects only the phonological well-formedness component of $F_{i,t}(u, c)$, not grammaticality.

LOCALISED EXCEPTIONS. Some derivational suffixes (e.g. *-imsi* ‘-ish’) are lexically marked “disharmonic”. The community memorizes them with $C_t(u, c) = 1$, so no feature remains unrealized and $G_t(u, c) = 1$ despite vowel mismatch. Clitic boundaries that start a fresh harmony cycle (Kabak & Vogel, 2001) are handled the same way: they satisfy the operator-exponent requirement and leave harmony to phonology alone.

In sum, Turkish suffix harmony errors are grammaticality failures because they mis-realize a closed inflectional operator exponent. Stem disharmony merely lowers phonological well-formedness and so affects only the listener’s feeling $F_{i,t}(u, c)$.

If it could be shown that phonology alone caused feelings of ungrammaticality, without realizing an operator contrast or expression-shape requirement, that would constitute a serious challenge to this framework’s central claim about the functional target of grammaticality.

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